

CHAPTER 3 : PROPERTIES OF PURE SUBSTANCE

Section	Core Revision Focus
3.1	Pure substance definition, fixed chemical composition, purity criterion, single/multiple components and phases
3.2	Solid/liquid/gas phases, molecular arrangement and motion, intermolecular forces, phase characteristics
3.3	Phase-change processes, compressed/saturated/superheated states, saturation temperature/pressure, latent heat, phase-change sequence
3.4	T-v, P-v, P-T diagrams, saturation dome, critical point, triple point, sublimation, phase regions, P-v-T surface
3.5	Property tables, saturated-liquid/vapor properties, quality, mixture-property relations, superheated vapor, compressed liquid, state identification
3.6	Ideal-gas equation $Pv = RT$, specific gas constant, molar form, two-state relation, constant-P/V/T cases, density/specific volume, ideal-gas applicability
3.7	Compressibility factor Z, deviation from ideal-gas behavior, reduced properties, principle of corresponding states, generalized compressibility chart
3.8	van der Waals, Beattie-Bridgeman, Benedict-Webb-Rubin, Strohbridge and virial equations, real-gas modeling, applicability and accuracy

3.1 PURE SUBSTANCE

Definition

A pure substance has a **fixed** chemical composition throughout.

Uniform chemical composition throughout

Key Criterion

A substance may contain one component, several components, or multiple phases and still be pure provided the composition is uniform.

Same composition in all regions/phases $\Rightarrow$ Pure substance

Different composition in different regions/phases $\Rightarrow$ Not a pure substance

Mixture	Pure Substance?
Ice + liquid water	Yes
Liquid air + gaseous air	No
Oil + water	No

Exam Point

- Number of components or phases alone does **not** determine purity.
- Multiple phases are permissible if all phases have the same chemical composition.

3.2 PHASES OF A PURE SUBSTANCE

Principal Phases

Solid, Liquid, Gas

Definition of a Phase

A phase has:

- a distinct molecular arrangement,
- a homogeneous structure, and
- identifiable boundaries separating it from other phases.

Molecular Comparison

Property	Solid	Liquid	Gas
Molecular arrangement	Ordered lattice	Less ordered	No order
Position	Relatively fixed	Not fixed	Widely separated
Motion	Oscillation	Rotation + translation	Random
Intermolecular force	Strongest	Intermediate	Weakest
Molecular energy	Lowest	Intermediate	Highest

Phase Transformation

$T \uparrow \Rightarrow$ Molecular velocity $\uparrow \Rightarrow$ Intermolecular forces can be overcome

Multiple Phases

The same substance may possess different phases within one principal phase because of different molecular structures.

Examples from the notes: graphite/diamond, multiple liquid phases of helium, multiple solid phases of iron, and multiple high-pressure ice phases.

3.3 PHASE-CHANGE PROCESSES OF PURE SUBSTANCES

Phase-Change Sequence

For constant-pressure heating of a pure substance:

Compressed liquid → Saturated liquid → Saturated mixture → Saturated vapor → Superheated vapor

Compressed Liquid / Subcooled Liquid

Liquid **not** about to vaporize.

Saturated Liquid: liquid **about** to vaporize.

Compressed liquid → Saturated liquid → Vaporization

Saturated Liquid–Vapor Mixture

During boiling:

- liquid and vapor coexist in equilibrium,
- heat transfer continues,
- temperature remains constant at constant pressure,
- liquid decreases while vapor increases,
- specific volume increases substantially.

Saturated liquid → Saturated mixture → Saturated vapor

Saturated Vapor

Vapor **about** to **condense**; any heat loss causes condensation.

Superheated Vapor

Vapor **not** about to condense.

At a specified pressure:

$T > T_{\text{sat}} \Rightarrow$ superheated vapor

Constant-Pressure Phase Change

During vaporization from saturated liquid to saturated vapor:

$P = \text{constant}, \quad T = \text{constant}$

For water at 1 atm:

$$T_{\text{sat}} \approx 100^{\circ}\text{C}$$

Water and Steam

Both refer to the same substance:



Saturation Temperature and Pressure

T_{sat} = temperature at which phase change occurs at a specified pressure

P_{sat} = pressure at which phase change occurs at a specified temperature

$$P_{\text{sat}} = f(T_{\text{sat}})$$

$$P_{\text{sat}} \uparrow \Rightarrow T_{\text{sat}} \uparrow$$

For water:

$$P_{\text{sat}} = 101.325 \text{ kPa} \Longleftrightarrow T_{\text{sat}} = 99.97^{\circ}\text{C}$$

Latent Heat

Energy absorbed or released during phase change.

Latent heat of fusion

$$h_{\text{fusion}} = 333.7 \text{ kJ/kg}$$

for water at 1 atm.

Melting: energy absorbed

Freezing: energy released

Latent heat of vaporization

$$h_{\text{vap}} = 2256.5 \text{ kJ/kg}$$

for water at 1 atm.

Vaporization: energy absorbed

Condensation: energy released

Practical Consequences

- Pressure increase $\Rightarrow$ boiling temperature increases.

- Elevation increase $\Rightarrow$ atmospheric pressure decreases $\Rightarrow$ boiling temperature decreases.
- Higher heat-transfer rate $\Rightarrow$ higher vaporization rate.
- Vacuum cooling uses evaporation and latent heat removal to lower product temperature.
- Vacuum freezing occurs when

$$P < 0.61 \text{ kPa}$$

for water at 0°C .

3.4 PROPERTY DIAGRAMS FOR PHASE-CHANGE PROCESSES

Important Diagrams

$$T - v, \quad P - v, \quad P - T$$

The complete three-dimensional representation is the $P-v-T$ surface.

$T-v$ Diagram

As pressure increases:

- T_{sat} increases,
- v_f increases,
- v_g decreases,
- distance between saturated-liquid and saturated-vapor states decreases.

Critical Point

At the critical point, saturated liquid and saturated vapor become identical.

Critical properties:

$$T_{\text{cr}}, \quad P_{\text{cr}}, \quad v_{\text{cr}}$$

Saturated liquid line + Saturated vapor line $\rightarrow$ Saturation dome

Region	Phase
Left of saturated-liquid line	Compressed liquid
Under saturation dome	Saturated liquid-vapor mixture
Right of saturated-vapor line	Superheated vapor

The left and right regions are **single-phase regions**; the region under the dome is a **two-phase equilibrium region** or **wet region**.

Supercritical Region

$$P > P_{\text{cr}}$$

No distinct liquid-vapor phase change occurs; properties vary continuously.

$P-v$ Diagram

The basic regions are the same as in the $T-v$ diagram.

During vaporization:

$$P = \text{constant}, \quad T = \text{constant}, \quad v \uparrow$$

Constant-temperature lines have a **downward trend**.

Solid Phase and Freezing

Most substances:

contract during freezing

Water:

expands during freezing

On a P-T diagram:

Behavior	Melting-Line Slope
Contracts on freezing	Positive
Expands on freezing	Negative

Triple-Phase Equilibrium

Solid + Liquid + Vapor

On P-v and T-v diagrams: **triple line**.

On P-T diagram: **triple point**.

For water:

$$T_{tp} = 0.01^{\circ}\text{C}$$

$$P_{tp} = 0.6117 \text{ kPa}$$

Thus:

$$T = 0.01^{\circ}\text{C}, \quad P = 0.6117 \text{ kPa}$$

P - v / T - v	Triple line
P - T	Triple point

Sublimation

Direct:

Solid $\rightarrow$ Vapor

Without passing through the liquid phase.

Common example: **dry ice** (CO_2).

Sublimation occurs at pressures below the triple-point pressure.

P-T Diagram / Phase Diagram

Line	Phases Separated
Sublimation line	Solid-Vapor
Melting line	Solid-Liquid
Vaporization line	Liquid-Vapor

Sublimation line $\cap$ Melting line $\cap$ Vaporization line = Triple point

Vaporization line ends at the critical point

Beyond the critical point there is no distinct liquid–vapor boundary.

P–v–T Surface

$$z = f(x, y)$$

For a simple compressible substance, **any two independent intensive properties** fix the state.

For the P–v–T surface:

- T and v may be independent,
- P is then dependent.

Every point on the surface represents an **equilibrium state**; a quasi-equilibrium process traces a sequence of such states.

Diagram Relationships

Diagram	Relation to P–v–T Surface
P – v	Projection on P – v plane
T – v	Bird's-eye view
P – T	Projection on P – T plane

Critical Point vs Triple Point

	Critical Point	Triple Point
Phases	Liquid + vapor become identical	Solid + liquid + vapor coexist
Location	End of vaporization line	Intersection of 3 phase lines
P – T	Point	Point
T – v	Point	Line
P – v	Point	Line

3.5 PROPERTY TABLES

Purpose

Property tables are used because relationships among thermodynamic properties are often too complex for simple equations.

Typical regions:

- saturated liquid / saturated vapor,
- saturated liquid–vapor mixture,
- superheated vapor,
- compressed liquid.

Enthalpy

$$h = u + Pv$$

$$H = U + PV$$

Therefore:

$$u = h - Pv$$

Saturation–Table Notation

- f → saturated liquid.
- g → saturated vapor.
- fg → vapor minus liquid.

$$v_{fg} = v_g - v_f$$

$$u_{fg} = u_g - u_f$$

$$h_{fg} = h_g - h_f$$

h_{fg} is the **enthalpy of vaporization** and decreases with increasing temperature/pressure, becoming zero at the critical point.

Quality

Quality x is the vapor mass fraction of a **saturated liquid–vapor mixture**:

$$x = \frac{m_g}{m_t}$$

$$m_t = m_f + m_g$$

$$0 \leq x \leq 1$$

$$x = 0 \Rightarrow \text{saturated liquid}$$

$$x = 1 \Rightarrow \text{saturated vapor}$$

Quality has meaning only for saturated liquid–vapor mixtures

Mixture Property Relations

$$v = (1 - x)v_f + xv_g$$

$$v = v_f + xv_{fg}$$

$$x = \frac{v - v_f}{v_{fg}}$$

$$u = u_f + xu_{fg}$$

$$h = h_f + xh_{fg}$$

General form:

$$y = y_f + xy_{fg} \quad y = v, u, h$$

Hence:

$$y_f \leq y \leq y_g$$

On $P - v$ or $T - v$ diagrams, quality is the distance ratio:

$$x = \frac{\text{distance from saturated liquid to actual state}}{\text{distance from saturated liquid to saturated vapor}}$$

Low $x \rightarrow$ near saturated liquid; high $x \rightarrow$ near saturated vapor.

Superheated Vapor

Single-phase vapor region.

$$P < P_{\text{sat}} \quad \text{at a given } T$$

$$T > T_{\text{sat}} \quad \text{at a given } P$$

$$v > v_g, \quad u > u_g, \quad h > h_g$$

Quality is **not defined**.

Compressed Liquid

When compressed-liquid data are unavailable, use the saturated-liquid approximation at the same temperature:

$$v \approx v_f@T$$

$$u \approx u_f@T$$

$$h \approx h_f@T$$

The approximation is generally good for v and u ; h is more pressure-sensitive.
Improved enthalpy approximation:

$$h \approx h_f@T + v_f@T (P - P_{\text{sat}}@T)$$

Characteristics:

$$P > P_{\text{sat}} \quad \text{at a given } T$$

$$T < T_{\text{sat}} \quad \text{at a given } P$$

$$v < v_f, \quad u < u_f, \quad h < h_f$$

Quality is **not defined**.

State Identification

Given P, T :

$$T < T_{\text{sat}}@P \Rightarrow \text{compressed liquid}$$

$$T = T_{\text{sat}}@P \Rightarrow \text{saturated state}$$

$$T > T_{\text{sat}}@P \Rightarrow \text{superheated vapor}$$

Alternatively, given T, P_{sat} :

$$P > P_{\text{sat}@T} \Rightarrow \text{compressed liquid}$$

$$P = P_{\text{sat}@T} \Rightarrow \text{saturated state}$$

$$P < P_{\text{sat}@T} \Rightarrow \text{superheated vapor}$$

Using a property $y = v, u, h$:

$$y < y_f \Rightarrow \text{compressed liquid}$$

$$y_f \leq y \leq y_g \Rightarrow \text{saturated liquid-vapor mixture}$$

$$y > y_g \Rightarrow \text{superheated vapor}$$

Reference State

Absolute values of u, h, s depend on the selected reference state; thermodynamic calculations generally use **property changes**.

$$\text{Use one consistent set of tables or charts}$$

Different table sets may have different absolute property values but give consistent property changes.

Table Selection

Known Information / Region	Table
Saturated state, T given	T -based saturation table
Saturated state, P given	P -based saturation table
Superheated vapor	Superheated-vapor table
Compressed liquid	Compressed-liquid table
Saturated solid-vapor mixture	Saturated solid-vapor table

3.6 THE IDEAL-GAS EQUATION OF STATE

Equation of State

An equation of state relates thermodynamic properties at equilibrium states; the basic form used here relates P , T , and v .

Experimental Basis

Boyle's law:

$$P \propto \frac{1}{V} \quad (T = \text{constant})$$

Charles/Gay-Lussac:

$$V \propto T \quad (P = \text{constant})$$

Ideal-Gas Equation

$$Pv = RT$$

where:

- P = absolute pressure,
- T = absolute temperature,
- v = specific volume,
- R = specific gas constant.

$$R = \frac{R_0}{M}$$

$$m = MN$$

R_0 = universal gas constant; M = molar mass; N = number of moles.

Alternative Forms

$$PV = mRT$$

$$PV = NR_0T$$

Molar specific volume:

$$\bar{v} = \frac{V}{N}$$

$$P\bar{v} = R_0T$$

Use bars for molar properties; unbarred quantities denote mass-based properties.

Two-State Relation

For fixed mass of ideal gas:

$$\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}$$

Special Cases

Constant volume

$$V = \text{constant}$$

$$\frac{P}{T} = \text{constant}$$

$$\frac{P_1}{T_1} = \frac{P_2}{T_2}$$

Constant pressure

$$P = \text{constant}$$

$$\frac{V}{T} = \text{constant}$$

$$\frac{V_1}{T_1} = \frac{V_2}{T_2}$$

Constant temperature

$$T = \text{constant}$$

$$PV = \text{constant}$$

$$P_1 V_1 = P_2 V_2$$

Ideal-Gas Applicability

Ideal gas = imaginary substance obeying:

$$Pv = RT$$

Real gases approach ideal behavior at **low density**, generally corresponding to:

- low pressure,
- high temperature.

Ideal-gas model is most appropriate at relatively low pressure and high temperature.
Dense gases, steam, and refrigerant vapor may require property tables.

Unit Requirements

Absolute pressure

$$P_{abs} = P_{gage} + P_{atm}$$

Absolute temperature

$$T(K) = T(^{\circ}C) + 273.15$$

Density and Specific Volume

Since:

$$v = \frac{1}{\rho}$$

$$P = \rho RT$$

$$\rho = \frac{P}{RT}$$

$$v = \frac{RT}{P}$$

Therefore, for fixed R:

$$\rho \uparrow \text{ with } P, \quad \rho \downarrow \text{ with } T$$

$$v \uparrow \text{ with } T, \quad v \downarrow \text{ with } P$$

3.7 COMPRESSIBILITY FACTOR---A MEASURE OF DEVIATION FROM IDEAL-GAS BEHAVIOR

Compressibility Factor

$$Z = \frac{Pv}{RT}$$

Thus:

$$Pv = ZRT$$

For an ideal gas:

$$Z = 1$$

Physical interpretation:

$$Z = \frac{v_{\text{actual}}}{v_{\text{ideal}}}$$

where:

$$v_{\text{ideal}} = \frac{RT}{P}$$

Therefore:

$$Z = 1 \Rightarrow \text{ideal-gas behavior}$$

$$Z > 1 \Rightarrow v_{\text{actual}} > v_{\text{ideal}}$$

$$Z < 1 \Rightarrow v_{\text{actual}} < v_{\text{ideal}}$$

The farther Z is from unity, the larger the deviation from ideal behavior.

Reduced Properties

$$P_R = \frac{P}{P_{cr}}$$

$$T_R = \frac{T}{T_{cr}}$$

Reduced properties are dimensionless.

Principle of corresponding states:

$$Z = f(P_R, T_R)$$

Different gases have approximately the same Z at the same P_R and T_R .

Generalized Compressibility Chart

The chart gives Z from P_R and T_R .

Once Z is known:

$$v = Z \frac{RT}{P}$$

or

$$v = Z v_{\text{ideal}}$$

Important Trends

At very low reduced pressure:

$$P_R \ll 1$$

$$Z \rightarrow 1$$

Hence:

$$P \rightarrow 0 \Rightarrow \text{real-gas behavior} \rightarrow \text{ideal-gas behavior}$$

At sufficiently high reduced temperature:

$$T_R > 2$$

ideal-gas behavior is generally a good approximation except at very high reduced pressures.

Near the critical point:

$$|Z - 1| \text{ can become large}$$

Thus real-gas deviation is especially important near the critical state.

Pseudo-Reduced Specific Volume

$$v_R = \frac{v_{\text{actual}}}{RT_{\text{cr}}/P_{\text{cr}}}$$

or

$$v_R = \frac{v P_{\text{cr}}}{RT_{\text{cr}}}$$

Used when P, v or T, v are known.

Determining v from P, T

1. Obtain $P_{\text{cr}}, T_{\text{cr}}$.

2. Calculate

$$P_R = \frac{P}{P_{\text{cr}}}, \quad T_R = \frac{T}{T_{\text{cr}}}.$$

3. Obtain Z from the generalized compressibility chart.

4. Calculate

$$v = Z \frac{RT}{P}.$$

Determining P from T, v

$$v_R = \frac{v P_{cr}}{RT_{cr}}$$

$$T_R = \frac{T}{T_{cr}}$$

Use v_R, T_R on the generalized chart to obtain P_R , then:

$$P = P_R P_{cr}$$

Accuracy

The generalized chart is approximate; accuracy depends on chart resolution, reading precision, and the extent to which the substance obeys the principle of corresponding states.

3.8 OTHER EQUATIONS OF STATE

General Point

These equations provide improved real-gas representation over a wider region than the ideal-gas equation.

Applicable to gas phase

They should **not** be used for liquids or liquid-vapor mixtures.

1. van der Waals Equation

$$\left(P + \frac{a}{v^2}\right)(v - b) = RT$$

Meaning:

- $\frac{a}{v^2}$ accounts for **intermolecular attraction**.
- b accounts for **molecular volume**.

Critical-point conditions:

$$\left(\frac{\partial P}{\partial v}\right)_{T=T_{cr}} = 0$$

$$\left(\frac{\partial^2 P}{\partial v^2}\right)_{T=T_{cr}} = 0$$

Constants:

$$a = \frac{27R^2T_{cr}^2}{64P_{cr}}$$

$$b = \frac{RT_{cr}}{8P_{cr}}$$

For molar form, use $\bar{v}$ and R_0 .

Limitation: conceptually useful and simple, but accuracy is often inadequate compared with more sophisticated equations.

2. Beattie-Bridgeman Equation

$$P = \frac{R_0 T}{\bar{v}^2} \left(1 - \frac{c}{\bar{v} T^3}\right) (\bar{v} + B) - \frac{A}{\bar{v}^2}$$

where:

$$A = A_0 \left(1 - \frac{a}{\bar{v}}\right)$$

$$B = B_0 \left(1 - \frac{b}{\bar{v}} \right)$$

Constants are experimentally determined and tabulated.

Applicability:

$$\rho \leq 0.8\rho_{cr}$$

More accurate than van der Waals over its useful density range.

3. Benedict–Webb–Rubin Equation

$$P = \frac{R_0 T}{\bar{v}} + \left(B_0 R_0 T - A_0 - \frac{C_0}{T^2} \right) \frac{1}{\bar{v}^2} + \frac{b R_0 T - a}{\bar{v}^3} + \frac{a\alpha}{\bar{v}^6} + \frac{c}{\bar{v}^3 T^2} \left(1 + \frac{\gamma}{\bar{v}^2} \right) e^{-\gamma/\bar{v}^2}$$

Constants:

$$A_0, B_0, C_0, a, b, c, \alpha, \gamma$$

determined experimentally.

Applicability:

$$\rho \leq 2.5\rho_{cr}$$

Handles substantially higher gas densities than Beattie–Bridgeman.

Strobridge Extension

Extension of Benedict–Webb–Rubin with more constants and greater flexibility.

Main use:

$$\text{digital computer calculations}$$

4. Virial Equation

$$P = \frac{RT}{\bar{v}} + \frac{a(T)}{\bar{v}^2} + \frac{b(T)}{\bar{v}^3} + \frac{c(T)}{\bar{v}^4} + \frac{d(T)}{\bar{v}^5} + \dots$$

Coefficients:

$$a(T), b(T), c(T), d(T), \dots$$

are **virial coefficients**, functions of temperature alone.

They may be obtained experimentally or through statistical mechanics.

More terms $\Rightarrow$ potentially greater accuracy.

At low pressure:

$$P \rightarrow 0$$

$$Pv \rightarrow RT$$

$$Z \rightarrow 1$$

Thus the virial equation is consistent with the ideal-gas limit.

Comparison

Equation	Constants	Accuracy	Use
van der Waals	Few	Limited	Simple real-gas modeling
Beattie-Bridgeman	Several	Reasonable	Wider gas-density range
Benedict-Webb-Rubin	More	High	Accurate real-gas calculations
Strobridge	Many	High	Computer-oriented calculations
Virial	Variable	Depends on terms	Series-based representation

Complexity vs Accuracy

$$\text{Greater complexity} \iff \text{generally greater accuracy and wider applicability}$$

- **van der Waals:** simplest; two constants; limited accuracy.
- **Beattie-Bridgeman:** more complex; experimentally determined constants.
- **Benedict-Webb-Rubin:** more constants; higher accuracy; wider density range.
- **Virial:** series form; accuracy increases with retained terms.

Calculation Consideration

These equations are generally **implicit in specific volume**; therefore solving for v may require **trial-and-error**.

Complex equations are particularly suitable for **digital computers**, while property tables or simpler equations are often more convenient for hand calculations.

Important Limitation

Do not use these gas-phase equations for liquids

Do not use these gas-phase equations for liquid-vapor mixtures

Use property tables or other appropriate relations for those regions.